Om Sharma

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B.Tech - Mechanical Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	8.30 (Current)	2025-Present
Senior Secondary (XII)	CBSE Board	86.2%	2025
Secondary (X)	CBSE Board	93.6%	2023

Projects

- **GradeOps: AI-Powered Human-in-the-Loop Exam Grading Platform** *Mar. 2025 - Present*
  Coding Club, IIT Guwahati · Python, FastAPI, React, LangGraph, Gemini, Nemotron, PostgreSQL Github
  – Built a platform that **OCRs handwritten exam scripts via NVIDIA Nemotron**, scores each answer against instructor rubrics using **Google Gemini**, and routes every result through a **human review queue** before grades are released.
  – Engineered a **LangGraph state machine** orchestrating the two-stage AI pipeline with **SQLite checkpointing** so pending reviews survive server restarts; backend served via **FastAPI on PostgreSQL** with role-based cookie auth.
  – Developed the **React frontend from scratch** – Instructor and TA dashboards, keyboard-friendly review queue, and a premium animated landing page with **Three.js** visuals.

- **Delhivery Network Intelligence: Graph-Based Logistics Optimization** *Jun. 2025*
  Consulting & Analytics Club, IIT Guwahati · Python, XGBoost, NetworkX, node2vec, Pandas, Streamlit Github
  – Modelled Delhivery's logistics network as a directed weighted graph on 144,900 scans to identify that 5 hubs account for 30.8% of total network excess delay; top bottleneck carries 12.5% of network delay alone.
  – Built graph-enhanced XGBoost ETA model using node2vec hub embeddings and betweenness centrality features, achieving **MAE 28 min vs OSRM's 108 min (74% improvement)** and 58.3% on-target accuracy vs 4.1%.
  – Delivered FTL vs Carting decision framework and upgrade impact analysis estimating **28% reduction in late deliveries** and **~Rs. 1.5 cr/yr** recovered revenue from top-3 hub upgrades.

- **Autonomous Bidding Agent for Used-Car Auctions** *Apr. 2025*
  Coding Club Recruitment, IIT Guwahati · Python, XGBoost, Pandas, Scikit-learn Github
  – Trained XGBoost regressor (log-target transform, engineered features: age, usage intensity, condition ratio) on historical US auction data; tuned via RandomizedSearchCV achieving **Val RMSE $1,620 – 14.3% improvement** over baseline.
  – Designed sigmoid-proportional bidding strategy with bankroll-adaptive ceiling and hard cap at 95% of predicted value, structurally guaranteeing profit on every win; deployed as a live auction simulation agent.

- **Bangladesh Power Demand Forecasting** *Apr. 2025*
  .ai Club, IIT Guwahati · Python, Scikit-learn, Random Forest, Pandas, NumPy Github
  – Fused PGCB power records, Dhaka weather observations, and World Bank economic indicators (~92,000 rows, 10-year span) into an end-to-end ML pipeline for hourly electricity demand forecasting.
  – Engineered autoregressive lag features (t-1h, t-24h, t-168h) and cyclical time encodings; tuned Random Forest via GridSearchCV with TimeSeriesSplit CV, achieving **MAPE $< 3\%$** and **$R^2 > 0.97$** on chronological hold-out.

Technical Skills

- **Programming Languages**: C/C++, Python
- **Libraries & Tools**: XGBoost, Scikit-learn, Pandas, NumPy, NetworkX, node2vec, Streamlit, Git, LaTeX
- **Key Concepts**: Machine Learning, Graph Analytics, Time Series Forecasting

Positions of Responsibility

- **Deputy Coordinator**, Coding Club, IIT Guwahati (Machine Learning Module) *Jun. 2026 - Present*
- **Associate**, Finance & Economics Club, IIT Guwahati *Jun. 2026 - Present*
- **Senior Executive**, Entrepreneurship Cell (E-Cell), IIT Guwahati *Oct. 2025 - Present*
- **Associate**, Aeromodelling Club, IIT Guwahati *Feb. 2026 - Present*
- **Associate**, Robotics Club, IIT Guwahati *Feb. 2026 - Present*

Key courses taken

- Programming and Data Structures, Linear Algebra, Multivariable Calculus, Real Analysis, Complex Analysis, Machine Learning Specialization (Andrew Ng)

Achievements

- **Codeforces**, Max Rating 1299 (Handle - om_sharma_), 200+ problems solved
  Greedy, Math, BS & DP *2025-Present*